

Dijkstra's algorithm

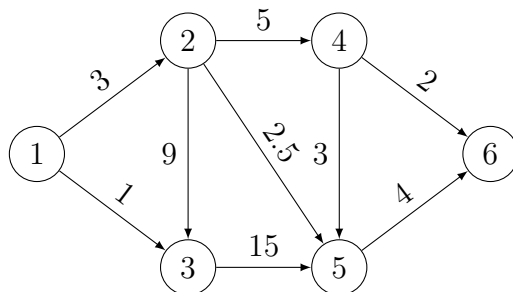
Michel Bierlaire

Practice quiz

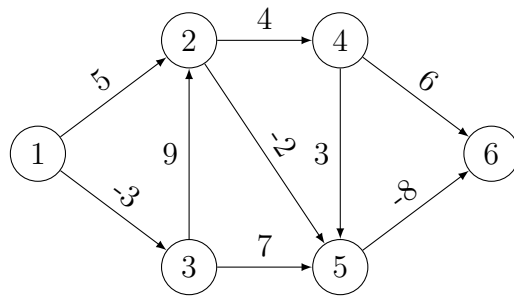
For each of the following networks, we consider the problem of finding the shortest path from node 1 to all other nodes.

- Is Dijkstra's algorithm applicable?
- If so, apply it to find the optimal labels and the shortest path spanning tree. Discuss the invariants of the algorithm during iteration 4.
- If not, apply the shortest path algorithm, where Dijkstra's rule for node selection is applied. It means that the node with the smallest label is selected to be processed at each iteration.

1. Network 1:



2. Network 2:



3. Network 3:

